Quadtratic Gravity in Regge Calculus

$f(R) = R + \alpha R^2$

Normal GR	Quadtatic GR
$I = \int R \underbrace{\sqrt{-g}\, d^4x}_{\text{Volume term}}$	$I = \int f(R) \sqrt{-g}\, d^4x$
	$= \int (R + \alpha R^2) \sqrt{-g}\, d^4x$
$\delta I = 0$	$\delta I = 0$

$\downarrow$ in Regge Calculus, Normal GR

$R = 2 \varepsilon_p \rho_p = 2 \varepsilon_p \dfrac{A_p}{V_p}$ ← 4-area (3-volume), ← 4-volume

$I \approx \sum R V_p \equiv \sum \varepsilon_p A_p$ (for all hinges p)

$$\frac{\delta I}{\delta l_p} = \underbrace{\sum \frac{\delta \varepsilon_p}{\delta l_p} A_p}_{0 \text{ (Schläfli}} + \boxed{\sum \varepsilon_p \frac{\delta A_p}{\delta l_p} = 0}$$

Quadratic Regge Calculus

$$I = \sum (R + \alpha R^2) V_p$$

$$I = \sum \left(2 \varepsilon_p A_p + \alpha \frac{4 \varepsilon_p^2 A_p^2}{V_p} \right)$$

$$\frac{\delta I}{\delta l_p} = \sum \Bigg(\underbrace{2 \frac{\delta \varepsilon_p}{\delta l_p} A_p}_{\to 0} + 2 \varepsilon_p \frac{\delta A_p}{\delta l_p} + \frac{8 \alpha \varepsilon_p A_p^2}{V_p} \frac{\delta \varepsilon_p}{\delta l_p} - \frac{4 \varepsilon_p^2 A_p^2}{V_p^2} \frac{\delta V_p}{\delta l_p} + \frac{8 \varepsilon_p^2 A_p}{V_p} \frac{\delta A_p}{\delta l_p} \Bigg) = 0$$

$$\boxed{\sum_p \left(\left(\varepsilon_p + \frac{4 \varepsilon_p^2 A_p}{V_p} \right) \frac{\delta A_p}{\delta l_p} + \frac{4 \alpha \varepsilon_p A_p^2}{V_p} \frac{\delta \varepsilon_p}{\delta l_p} - \frac{4 \varepsilon_p^2 A_p^2}{V_p^2} \frac{\delta V_p}{\delta l_p} \right) = 0}$$

In terms of $\varsigma_p = \ell_p^2$

same thing:

$$\sum_p \left(\left(\varepsilon_p + \frac{4 \varepsilon_p^2 A_p}{V_p} \right) \frac{\delta A_p}{\delta \varsigma_p} + \frac{4 \alpha \varepsilon_p A_p^2}{V_p} \frac{\delta \varepsilon_p}{\delta \varsigma_p} - \frac{4 \varepsilon_p^2 A_p^2}{V_p^2} \frac{\delta V_p}{\delta \varsigma_p} \right) = 0$$

$$\frac{\delta}{\delta x} \doteq \frac{d}{dX}$$

In 3+1 Decomposition, No Quadratic

$$A_{ijk} = \frac{1}{4}\sqrt{\left|2(s_{ij}s_{ik} + s_{ij}s_{kj} + s_{jk}s_{ik}) - s_{ij}^2 - s_{ik}^2 - s_{jk}^2\right|}$$

$$A_{ijk} = \frac{1}{4}\sqrt{|Q_{ijk}|}$$

$$s = l^2$$

$$\varepsilon_{ijk} = \begin{cases} 2\pi - \sum_{\sigma \supset (ij,k)} \theta_{ijk}^{\sigma}, & Q_{ijk} > 0 \\ \sum_{\sigma \supset (ij,k)} \theta_{ijk}^{\sigma}, & Q_{ijk} < 0 \end{cases}$$

sign change integrated from change of sign of derivative!!!

$$g_{ab} = \frac{1}{2}(s_{0a} + s_{0b} - s_{ab})$$

inverse

$$\cos\theta_{ijk}^{\sigma} = -\frac{g^{lm}}{\sqrt{g^{ll}g^{mm}}} \rightarrow \text{arccos or arcosh}$$

$i,j,k,l,m \to 0,1,2,3,4$

$1,2,3,4 \supset (a,b)$

$$\frac{\delta A_{ijk}}{\delta s_{ij}} = \frac{s_{ik} + s_{jk} - s_{ij}}{16 A_{ijk}}$$

$$R_{ij} = \sum_k \varepsilon_{ijk} \frac{\delta A_{ijk}}{\delta s_{ij}} = 0 \quad \text{Normal GR}$$

Quadratic GR Regge
3+1

$$\varepsilon_{ijk} = \begin{cases} 2\pi - \sum_{\sigma \supset (i,j,k)} \theta^\sigma_{ijk}, & Q_{ijk} > 0 \\ -\sum_{\sigma \supset (i,j,k)} \theta^\sigma_{ijk}, & Q_{ijk} < 0 \quad !!! \end{cases}$$

$$\frac{\delta A_{ijk}}{\delta s_{ij}} = \mathrm{sgn}(Q_{ijk}) \frac{s_{ik} + s_{jk} - s_{ij}}{16 A_{ijk}}$$

Removing absorbed minus due to stranger terms in quadratic formulation

Could use Brewin deficit derivative formula, but might be too complicated? $m^\mu = ?$, $n^\mu = ?$

$$\frac{\delta \theta^\sigma_{ijk}}{\delta s_{ij}} = \frac{d}{ds_{ij}} \left(\frac{-g^{lm}}{\sqrt{g^{ll} g^{mm}}} \right) \rightarrow \text{algebraic manipulation}$$

ordered again

$$\frac{\delta \varepsilon_{ijk}}{\delta s_{ijk}} = -\sum_\sigma \frac{\delta \theta^\sigma_{ijk}}{\delta s_{ij}}$$

$$\frac{dg^{ab}}{ds_{ij}} = \sum_{\text{sum}} g^{ac} \frac{dg_{cd}}{ds_{ij}} g^{db} \qquad c,d \to 1,2,3,4 \quad (?)$$

$$\frac{dg_{ab}}{ds_{ij}} = \frac{1}{2} \left(\frac{ds_{0a}}{ds_{ij}} + \frac{ds_{0b}}{ds_{ij}} - \frac{ds_{ab}}{ds_{ij}} \right)$$

$$\pm\frac{1}{2}, 0, 1 \qquad 0,1 \qquad 0,1 \qquad 0,1$$

$$\frac{\delta \theta^\sigma_{ijk}}{\delta s_{ij}} = \boxed{\frac{-1}{\sqrt{1 - \cos^2 \theta^\sigma_{ijk}}}} \left(-\frac{\frac{\delta g^{lm}}{\delta s_{ij}}}{\sqrt{g^{ll} g^{mm}}} + \frac{g^{lm}}{2 (g^{ll} g^{mm})^{3/2}} \left(\frac{\delta g^{ll}}{\delta s_{ij}} g^{mm} + \frac{\delta g^{mm}}{\delta s_{ij}} g^{ll} \right) \right)$$

changes sign depending on if $\cos\theta > \text{or} < 1$

$V \sim \sqrt{\det|g|}$

of one simplex σ:

$$V_\sigma = \frac{1}{4!} \sqrt{\det|g_{ab}|} \quad \text{in 4-d}$$

(Marco Berin, 2011)

$$\frac{dV_\sigma}{ds_{ij}} = \frac{1}{24} \frac{1}{2\sqrt{\det|g_{ab}|}} \frac{d(\det|g_{ab}|)}{ds_{ij}}$$

$$= \frac{1}{24} \frac{1}{2\sqrt{\det|g_{ab}|}} \det|g_{ab}| \, \mathrm{tr}\left(g^{ab} \frac{dg_{ab}}{ds_{ij}} \right)$$

$$= \frac{1}{24} \frac{\sqrt{\det|g_{ab}|}}{2} \mathrm{tr}\left(g^{ab} \frac{dg_{ab}}{ds_{ij}} \right)$$

$$= \frac{V_\sigma}{2} \mathrm{tr}\left(g^{ab} \frac{dg_{ab}}{ds_{ij}} \right)$$

we need the volumes associated with the triangles

σ has 5 vertecies

$\Rightarrow \frac{5!}{3!\,2!} = 10$ triangles

$$\Rightarrow V_p = \frac{V_\sigma}{10}$$

$$\frac{dV_p}{ds} = \frac{1}{10} \frac{dV_\sigma}{ds}$$